

23-2-2018

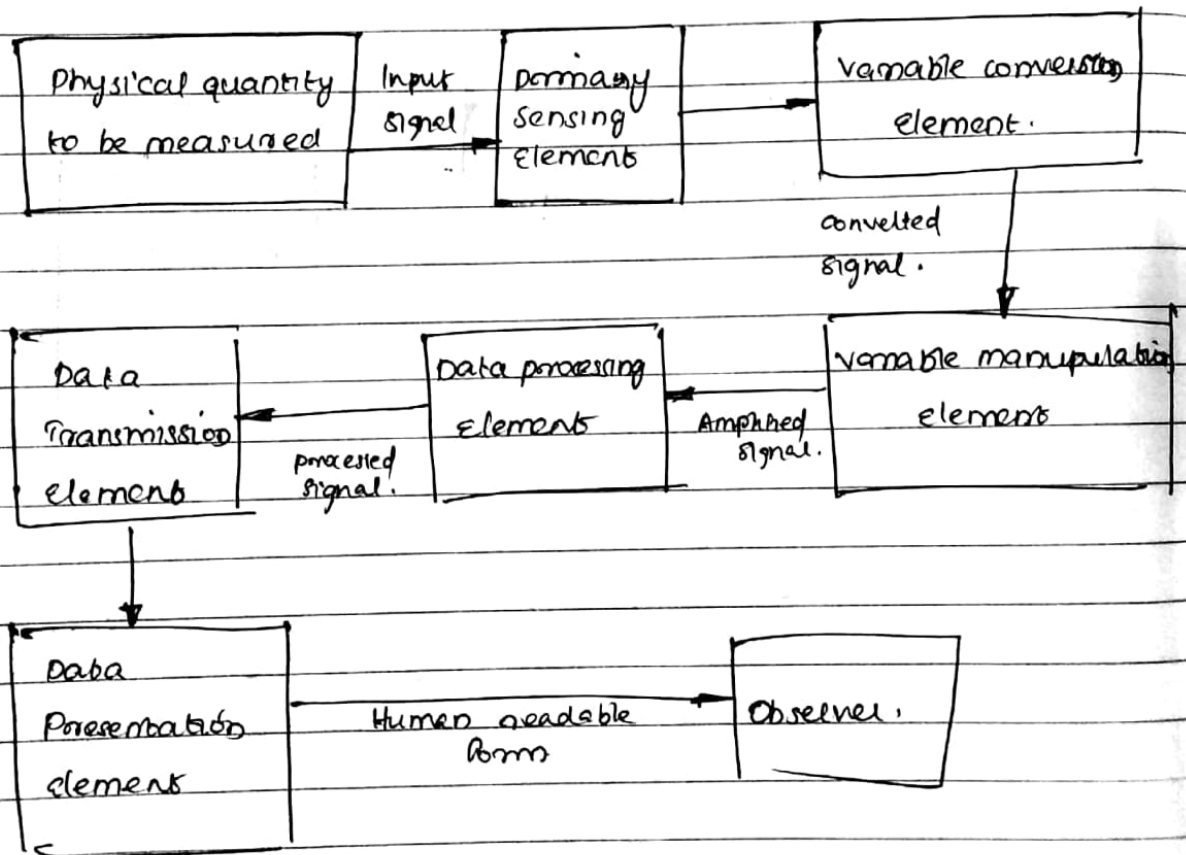
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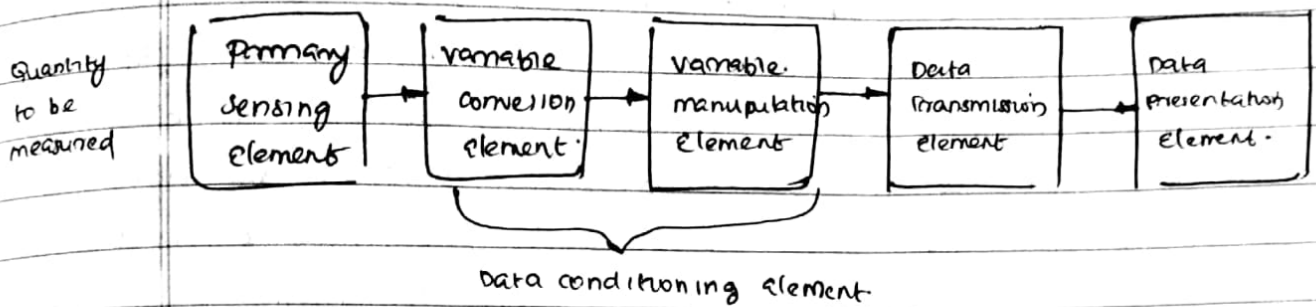
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MODULE - III

APPLICATIONS OF MEASUREMENT SYSTEM

Generalized Measurement System.





Generalized Measurement System.

It consists of the following components.

1. Primary sensing element
2. Variable conversion element
3. Variable manipulation element
4. Data transmission system
5. Data processing element
6. Data Presentation element

1. Primary sensing element.

It receives signal of the physical quantity to be measured as input. It converts the signal to a suitable form so that it becomes easier for other elements of the measurement system to either convert or manipulate.

2. Variable conversion

It converts the output electrical/mech. signal of primary sensing element which may be voltage, frequency, etc into a more suitable form without changing the information content of the input signal. (In some instruments there is no need of using variable conversion elements)

3. variable manipulation element

It manipulates & amplifies the output of the variable conversion element, for eg: an electronic voltage amplifier receives a small voltage as input and produces a greater magnitude of volt as output. It also removes noise from the system.

4. Data Processing element

It is used to modify the data before displayed or finally recorded. It may provide corrections to the measured physical variable to compensate for zero offset, temp element, scaling etc.

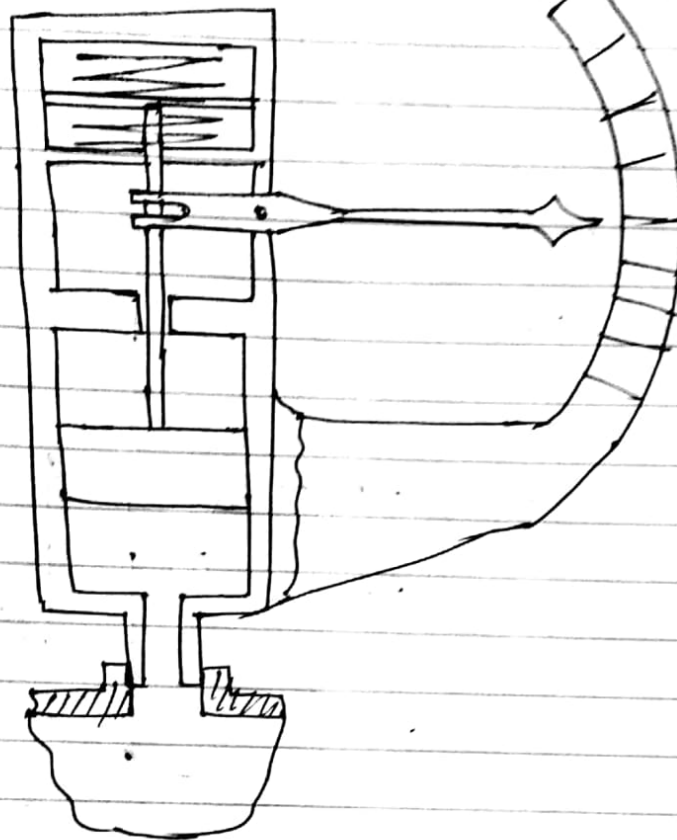
5. Data Transmission system

It is used for transmitting data from one element to another. It acts as a communication link b/w different elements of measurement system. Some of these data transmission elements are cables, wires, antenna, fibre etc.

6. Data Presentation element

It is used to present the measured physical quantity in human readable form to the observer. These are the elements that finally communicate the information of the measured variable to a human observer for monitoring, control or analysis purpose. The values of the measured variable may be indicated by analogue indicators (pointer & scale), digital indicators (ammeter/voltmeter) or recorder (camera, TV equipment etc).

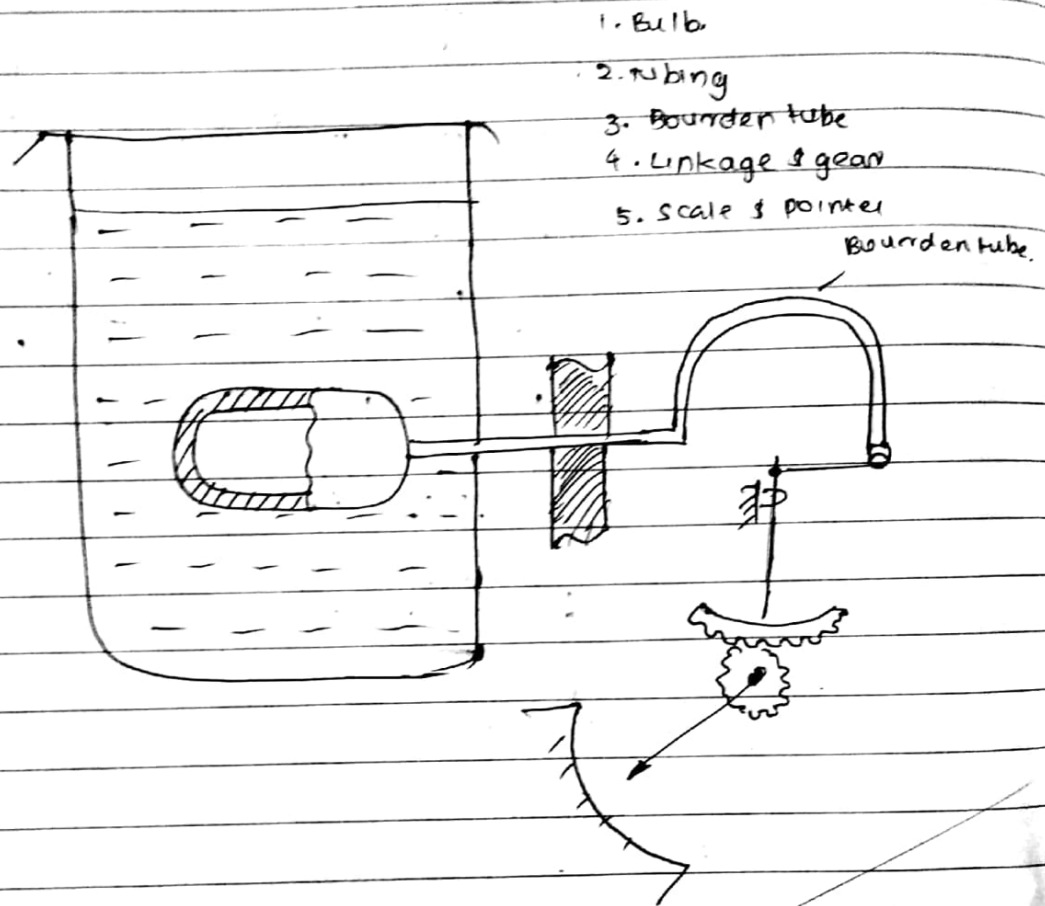
Function element of Pressure Gauge.



As an example of above concepts consider a pressure gauge.

- Primary sensing element is the piston, which also serves the function of variable conversion, since it converts fluid pressure to a resultant force on the piston face.
- Force is transmitted by the piston rod to the spring which converts the force to a proportional displacement. This displacement of the piston rod is magnified (multiplied) by the linkage to give a larger pointer displacement.
- The pointer & scale indicate the pressure thus serving as data presentation element.
- If it were necessary to locate the gauge at some distance from the source of pressure, a small tube would serve as data transmission element.

Functional Element of Pressure Thermometer.



The above fig shows a pressure type thermometer. The liquid filled bulb acts as a primary sensor and a variable conversion element. Temp change results in a pressure diff build up in the bulb. Because of the constrained thermal expansion of filling tube. This pr is transmitted through the tube to a Bourdon type pr gauge which convert pr to displacement. This displacement is manipulated by a linkage & gearing to a larger pointer motion. Scaled pointer ~~comes~~ again results in data presentation.

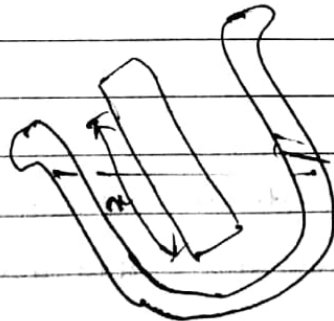
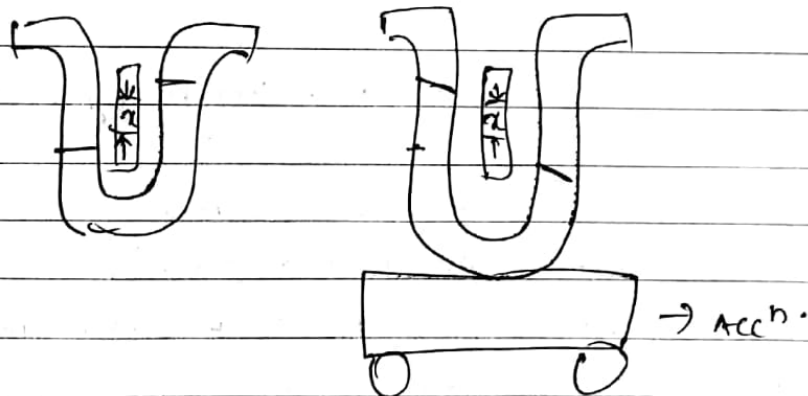
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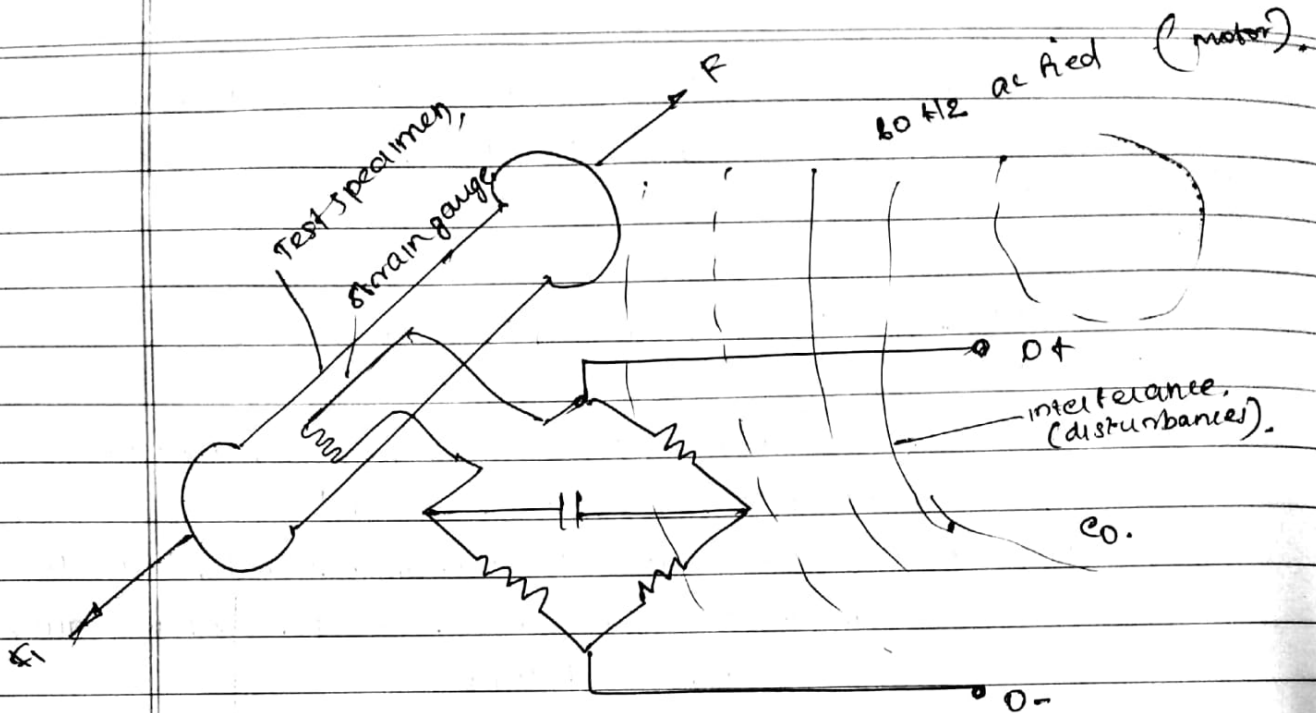
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INPUT OUTPUT CONFIGURATION OF INSTRUMENTS.

Input quantities are classified into 3.

- ^{input} Desired ~~output~~ → measurands or quantities that the instrument is designed to be measured
- Interfering input → quantities that unintentionally affect the instrument as consequence of the principles used to acquire and process the desired input.
- Modifying input → undesired quantities that affect the.





Desired Input $\rightarrow$ Temp.

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NB

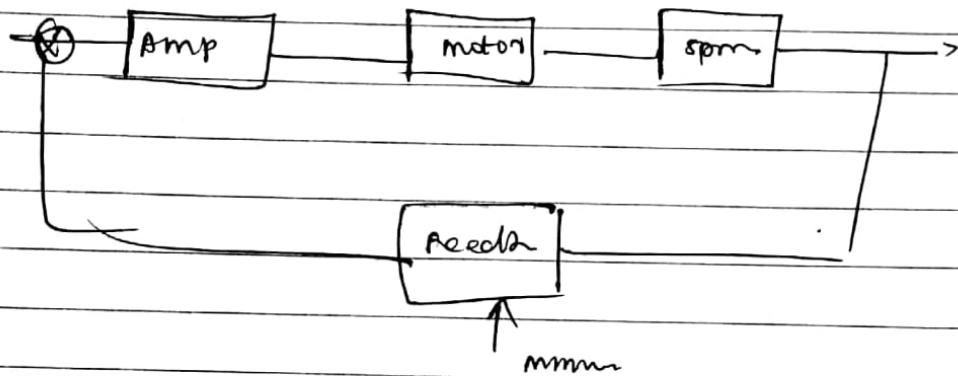
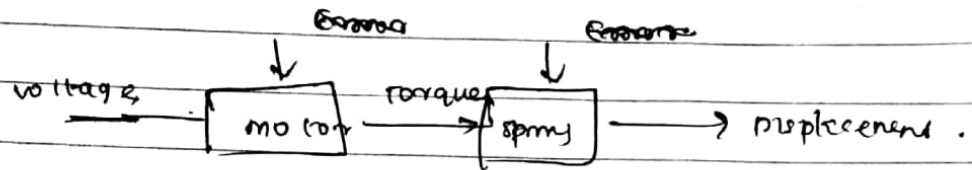
Methods of Connection for Interfering and Modifying Input.

1. Method of inherent insensitivity
2. Method of high gain feed back.
3. Method of calculated output connections.
4. Method of signal filtering.
5. Method of opposing input

1.

- $\rightarrow$ sound design $\rightarrow$ sensitive to desired outputs only.
- $\rightarrow$ In strain gauge use a gauge material of low temp co-efficient of resistance.
- $\rightarrow$ mech apparatus $\rightarrow$ use. low temp co-efficient of expansion/contraction.
- $\rightarrow$ If stiffness must be temp insensitive, consider nickel iron-chromium only.

2. Method of high gain feedback.



3-

4.

filtering unwanted signals. before affecting the system.

5.

